UNIVERSITÀ DI PISA

Distributed Data Mining and Analysis

Project report

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Chapter 1

Introduction

This report presents a distributed data mining analysis project focused on the Amazon Books Reviews dataset. The project relies on two main datasets: a metadata dataset describing books and a review-level dataset containing user ratings and textual reviews. Together, these datasets combine structured attributes such as price, publication year, and genre with unstructured textual content from user reviews.

The primary objective of this project is to investigate the determinants of customer satisfaction at scale. Specifically, we examine how book price, genre, popularity, and linguistic patterns in reviews relate to average ratings and sentiment polarity. Given the substantial size of the dataset, distributed data processing techniques implemented in Apache Spark were employed to ensure scalability, computational efficiency, and reproducibility.

Overall, the project demonstrates how distributed data mining methods can be applied to large real-world datasets to extract meaningful insights from both numerical and textual information. This project follows the CRISP-DM methodology: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment considerations.

Chapter 2

Data Understanding and Preparation

2.1 Dataset Description

The metadata dataset contains approximately 212,000 unique books retrieved via the Google Books API. The review dataset includes nearly 3 million user-generated reviews.

The metadata dataset includes attributes such as title, authors, publisher, publication date, categories, price, and ratings count. The review dataset contains review-level information including user ID, review score, review time, summary, and full review text.

2.2 Data Cleaning and Preprocessing

2.2.1 Missing Value

A systematic null-value assessment revealed substantial missingness in several metadata attributes. In particular, *ratingsCount* exhibited approximately 70% missing values, while *publisher* contained approximately 36% missing entries. Given the high sparsity of *ratingsCount* and the absence of a reliable imputation strategy, this variable was excluded from predictive modeling to prevent the introduction of artificial signal or bias.

Additionally, link-related attributes such as *previewLink* and *infoLink* were removed, as they do not contribute to the analytical objectives of this study and provide no predictive value.

In the review dataset, missing values were minimal relative to total volume. Records lacking *review/text* were removed due to their essential role in NLP tasks. Non-informative variables such as *profileName* were discarded and a small number of missing *Price* entries were removed or filtered during numeric casting.

2.2.2 Rating Cleaning

The review/score column contained valid floats, fractional formats, and textual noise. Regular expressions were applied to extract valid ratings in the range [1,5]. Invalid entries were discarded, and the column was cast to numeric.

2.2.3 Price Cleaning

The *Price* column exhibited several structural inconsistencies, such as numeric values stored as strings, missing entries, occasional malformed or non-numeric values.

To ensure numerical consistency and analytical reliability, the following preprocessing steps were applied such as, regular expression filtering to retain valid numeric values only, removal of non-positive price entries, casting of the cleaned column to `Double` type.

After cleaning, price values ranged from 1 USD to 995 USD, with the majority of observations concentrated below 50 USD, indicating a right-skewed distribution. Extreme values were retained and therefore represent meaningful market variability rather than data errors.

2.2.4 Text Preprocessing

Text preprocessing consisted of several normalization and filtering steps to ensure data consistency and modeling suitability. Specifically:

- Conversion of all text to lowercase,
- Removal of empty or null entries,
- Encoding normalization to ensure character consistency,
- Filtering of extremely short reviews lacking sufficient semantic content.

From an engineering perspective, all preprocessing steps were implemented using distributed Spark DataFrame operations, including `withColumn()`, `filter()`, `drop()`, `split()`, and `cast()`. Importantly, large `collect()` operations were explicitly avoided in order to preserve distributed scalability across approximately three million review records.

Intermediate DataFrames were cached selectively to optimize execution efficiency and reduce redundant recomputation within the Spark execution DAG. This strategy improved performance while maintaining memory stability.

As a result, the final prepared dataset is numerically consistent, structurally coherent, and fully compatible with distributed processing workflows. It provides a robust foundation for subsequent exploratory analysis, supervised classification, and unsupervised modeling tasks.

2.3 Exploratory Data Analysis

2.3.1 Rating Distribution Analysis

The distribution of review scores exhibits strong positive skewness, with the majority of observations concentrated in the upper rating categories. Specifically, most reviews are rated 4 or 5 stars, yielding an overall average rating of approximately 4.2.

This pronounced positivity bias suggests that users are more likely to submit reviews for books they enjoyed, a common phenomenon in online review systems. Such bias must be considered when interpreting subsequent statistical analyses and modeling results.

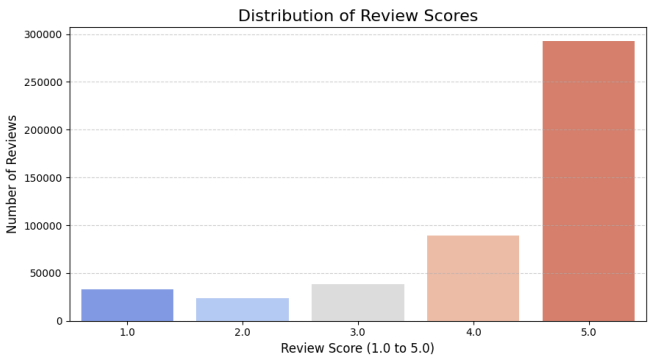

Figure 2.1: Distribution of review scores.

The bar chart visualization:

- 5-star reviews dominate the dataset,
- 4-star reviews are also highly frequent,
- 1-star and 2-star reviews are comparatively rare.

This distribution implies a concentration of variance within the upper rating categories and a limited representativeness of negative sentiment. From a statistical perspective, skewed label distributions increase the

risk of inflated accuracy metrics, particularly in supervised learning settings. Models trained on imbalanced data may become biased toward majority classes, thereby overstating predictive performance if minority-class behavior is not properly addressed through appropriate evaluation metrics or balancing strategies.

2.3.2 Price Distribution and Price–Rating Relationship

Book prices range from 1 USD to 995 USD, with an average price of approximately 21.9 USD. The majority of observations are concentrated below 50 USD, indicating a right-skewed distribution. This pattern reflects the typical retail structure of the book market, where standard editions are relatively affordable, while special, collector, or academic editions may command substantially higher prices.

Statistically, the Pearson correlation coefficient between price and rating is close to zero, indicating negligible linear dependence. These findings suggest that perceived quality, as reflected in user ratings, is largely independent of monetary cost.

2.3.3 Genre Distribution Analysis

Genre analysis reveals substantial imbalance:

- Fiction accounts for approximately 31% of books.
- Religion and History are also well represented.
- Science and Social Science appear significantly less frequently.

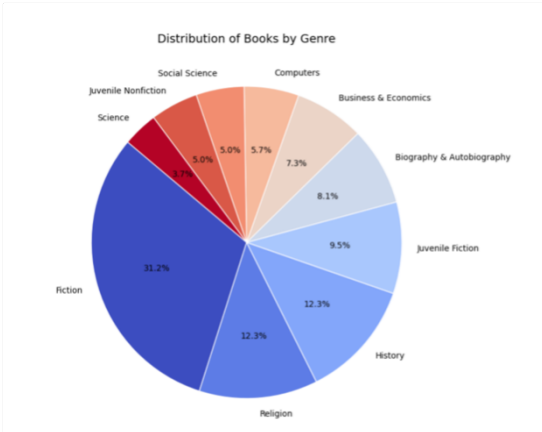

Figure 2.2: Distribution of books by genre.

The dominance of fiction may influence aggregate statistics and bias modeling results toward popular genres. This imbalance impacts aggregate statistics, as dominant genres exert greater influence on average rating calculations and clustering outcomes.

2.3.4 Distribution by Publication Year

Temporal analysis of publication year demonstrates a pronounced increase in the number of books in recent decades. This growth is likely attributable to digitization, expanded publishing platforms, and improved metadata availability.

However, this temporal concentration may introduce bias in review behavior and genre popularity, as newer publications may benefit from increased online visibility and engagement compared to older works.

2.3.5 Textual Exploration

Most Frequent Words in Reviews

Token frequency analysis of review text reveals recurrent terms associated with narrative structure and reader evaluation. The most common words include *story*, *like*, *read*, *good*, and *great*.

These terms reflect subjective assessment and engagement with narrative elements, indicating that readers primarily evaluate books based on storytelling quality and personal enjoyment.

Most Frequent Words in Titles

Frequency analysis of book titles highlights recurring thematic patterns, including *series*, *edition*, *world*, *american*, and *war*.

These terms suggest the prominence of serialized publications, historical narratives, and culturally contextualized themes within the dataset. The presence of such recurring keywords reflects broader publishing trends and thematic concentrations in contemporary literature.

2.3.6 Review Length–Rating Relationship

To investigate whether sentiment intensity influences textual verbosity, review length was analyzed across rating categories. As shown in Figure 2.3, lower-rated reviews (particularly 1- and 2-star ratings) exhibit slightly higher median lengths and greater dispersion compared to 5-star reviews.

This pattern suggests that dissatisfaction may encourage users to provide more detailed explanations of perceived shortcomings, whereas highly positive reviews tend to consist of shorter, more concise affirmations.

However, review length is not exclusively determined by rating. Substantial dispersion is observed within each category, indicating that verbosity varies considerably across individual users and contexts.

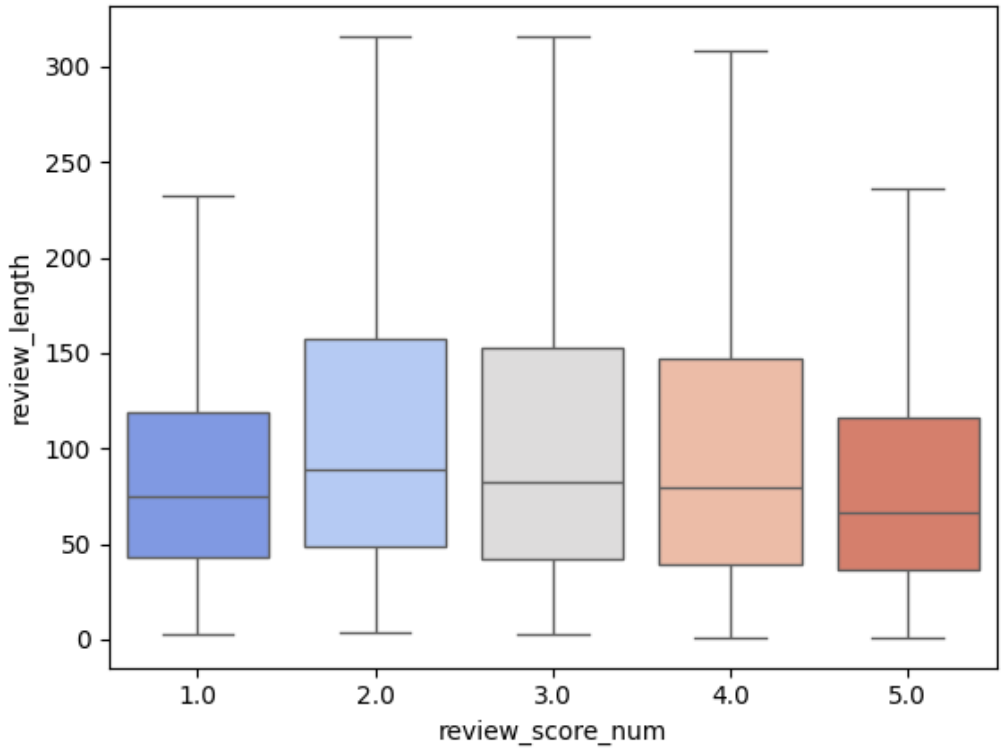

Figure 2.3: Distribution of review length across rating categories. Lower-rated reviews show slightly higher median length and dispersion.

Chapter 3

Modeling and Predictive Analysis

3.0.1 Research Question 1: Does Review Volume Predict Rating?

To investigate whether review volume influences perceived quality, reviews were aggregated at the book level by computing total review count and average rating per title. The Pearson correlation between review count and average rating was approximately zero, indicating no meaningful linear association.

In contrast, the Spearman rank correlation revealed a weak negative monotonic relationship ($\rho \approx -0.22$). This pattern reflects statistical stabilization rather than a causal relationship between popularity and quality. Books with few reviews exhibit higher variance in average ratings due to limited sample size, whereas books with many reviews converge toward a population mean.

3.0.2 Research Question 2: Do More Expensive Books Receive Higher Ratings?

We wanted to check whether price influences perceived quality, correlations between book price and average rating were computed. Both Pearson and Spearman coefficients were negligible, indicating neither linear nor monotonic dependence.
These findings suggest that consumer satisfaction, as measured by ratings, is not systematically associated with monetary cost. Price appears to reflect market positioning rather than intrinsic user-perceived quality.

Figure 3.1: Relationship between book price and average rating.

3.0.3 Research Question 3: Which Genres Tend to Have the Highest Average Ratings?

Average ratings were calculated for each genre after book-level review data was aggregated and combined with metadata that included categorical designations in order to assess whether genre affects perceived quality.

Although minor differences in average ratings across genres were observed, these variations are relatively small when compared to the overall positivity bias of the dataset. No genre exhibits a substantially higher or lower rating distribution that would suggest a structurally determined quality effect.

These findings indicate that genre alone does not strongly determine user satisfaction. While small genre-level patterns may exist, they do not represent a dominant explanatory factor in rating behavior.

3.0.4 Research Question 4: Is Linguistic Intensity Associated with Extreme Ratings?

To examine whether linguistic intensity correlates with extreme sentiment, 1-star and 5-star reviews were compared. Emotional intensity in online text is often expressed through stylistic features such as exclamation marks, question marks, and capitalization patterns. Psychological literature suggests that both positive enthusiasm and negative frustration may produce stronger expressive signals.

Table 3.1: Linguistic intensity metrics for extreme ratings (1-star vs 5-star).

Rating	Avg Exclam./100w	Avg Quest./100w	Avg Caps Ratio	Avg ALLCAPS/100w	Avg Words
1.0	0.74	0.334	0.043	1.042	101.7
5.0	0.96	0.124	0.042	0.832	101.9

The analysis reveals systematic differences between extreme rating groups. Five-star reviews exhibit a higher frequency of exclamation marks, indicating enthusiastic and affirmative expressions. In contrast, one-star reviews contain more question marks and all-capitalized words, reflecting skepticism, emphasis, or dissatisfaction.

However, the overall capitalization ratio does not differ substantially between the two groups. This suggests that specific stylistic markers, rather than general capitalization intensity, are associated with extreme sentiment expression.

3.1 Unsupervised Learning: Clustering Analysis

An unsupervised learning approach was applied to explore the latent structure of the review corpus without relying on sentiment labels. The objective was to determine whether reviews naturally form meaningful groups based on linguistic patterns, writing style, or structural characteristics.

Text Representation

Before clustering, review text was transformed into numerical features using a distributed Spark ML pipeline. The preprocessing workflow included tokenization, stopword removal, term-frequency computation using HashingTF, and inverse document frequency weighting (TF–IDF).

Given the large size of the dataset, clustering was performed on a 20% sample of the processed sentiment dataset to reduce computational cost while preserving representativeness of the corpus.

3.1.1 K-Means Clustering

K-means clustering on TF–IDF features was selected as a simple and computationally efficient approach for large-scale text segmentation. TF–IDF provides a sparse representation that emphasizes discriminative vocabulary, while K-means documents according to centroid-based similarity in this high-dimensional space. K-Means is commonly applied to TF–IDF representations of text documents because cosine or Euclidean distances between sparse vectors capture lexical similarity between documents. The algorithm partitions the dataset into k clusters by minimizing the within-cluster sum of squared distances.

Selection of the Number of Clusters

To determine an appropriate number of clusters, models were evaluated for $k = 2, 3, 4, 5$. The corresponding silhouette scores were:

Figure 3.2: Silhouette score visualization for different values of k.

Number of Clusters	Silhouette Score
2	0.5722
3	0.5690
4	0.3567
5	0.3562

Table 3.2: Silhouette scores for different numbers of clusters

Although $k = 2$ produced the highest score, the difference between $k = 2$ and $k = 3$ was small. The configuration $k = 3$ was therefore selected because it provided more interpretable segmentation of review behavior while maintaining strong cluster separation.

Cluster Profiling

To interpret the clusters, descriptive statistics were computed using review length measured in characters. The results reveal significant differences between clusters:

Cluster	Average Review Length	Interpretation
0	∼422 characters	Short, concise reviews
1	∼1854 characters	Long analytical reviews
2	∼11,811 characters	Extremely long records (likely anomalies)

Cluster 0 represents the vast majority of the dataset (over 470,000 reviews) and consists primarily of short, subjective opinions with an average length of approximately 422 characters. These reviews typically express direct personal evaluations of a book.

Cluster 1 forms a distinct group of approximately 71,000 records containing substantially longer reviews (average length of about 1,854 characters). These texts often resemble analytical descriptions or extended summaries that focus on plot elements, historical context, or thematic discussion.

Cluster 2 contains only 24 records but is characterized by extremely long texts (average length of approximately 11,811 characters). Qualitative inspection revealed that these records consist largely of catalog metadata or publisher descriptions rather than genuine user-generated reviews. This demonstrates that the clustering approach was able to isolate anomalous entries that would ideally be removed during data cleaning.

Cluster Visualization

To visualize the cluster structure, Principal Component Analysis (PCA) was applied to reduce the TF–IDF feature space to two dimensions. Although dimensionality reduction inevitably compresses information, the PCA projection allows a qualitative inspection of the separation between clusters.

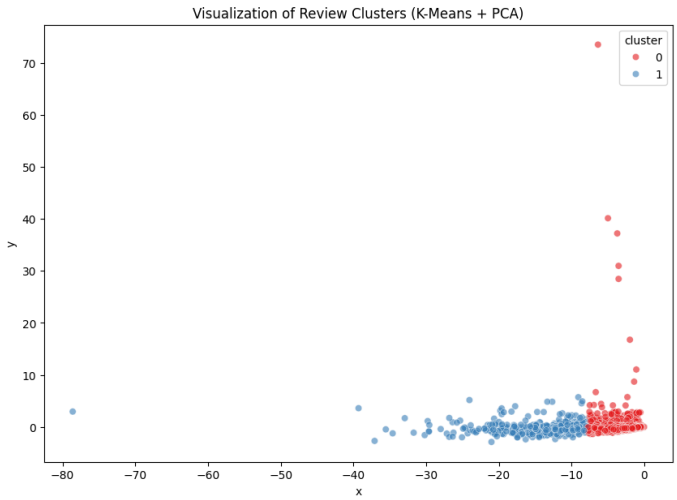

Figure 3.3: PCA projection of TF–IDF features colored by K-Means cluster assignment.

3.1.2 Conclusion

As part of the exploratory data analysis (EDA) and feature engineering process, K-Means clustering was applied to TF–IDF representations of the review texts. Principal Component Analysis (PCA) was subsequently used to visualize the high-dimensional structure of the data. The goal of this analysis was to uncover latent patterns in the review corpus without relying on sentiment labels.

The results, supported by a relatively high Silhouette Score of approximately 0.57 and the PCA visualization, indicate that the dataset naturally separates into distinct groups based on linguistic style and textual structure. Profiling the clusters using review length revealed three clearly interpretable segments. Cluster 0 represents typical consumer reviews, which tend to be relatively short and focused on subjective evaluation. Cluster 1 contains longer and more elaborate reviews that resemble detailed critiques or summaries. Cluster 2 contains unusually long texts that are unlikely to represent genuine user reviews. These records likely correspond to copied metadata, publisher descriptions, or other forms of noisy data.

The clustering results are also relevant for interpreting the subsequent sentiment classification task. In particular, the separation between short emotional reviews, longer analytical reviews, and anomalously long textual records suggests that the corpus is not homogeneous in writing style. This heterogeneity helps explain why simple sentiment models may struggle in some cases. Short, strongly opinionated reviews often contain direct lexical cues of polarity, whereas long analytical reviews may mix positive and negative expressions, discuss plot or historical context, and use more neutral vocabulary overall. As a result, reviews with similar ratings may differ substantially in linguistic structure, making linear sentiment separation more difficult. The clustering analysis therefore provides an important bridge between exploratory analysis and supervised learning, showing that classification performance is influenced not only by sentiment polarity itself but also by differences in writing style and review composition.

3.2 Supervised Learning: Sentiment Classification

Sentiment analysis was conducted on book reviews contained in the `ratings_df` DataFrame. The objective was to predict whether a review expresses positive sentiment (review score ≥ 4) or negative sentiment (review score ≤ 2) based on the textual content of the `review/text` column. Neutral reviews (rating $= 3$) were excluded to increase class separability and reduce ambiguity in sentiment labeling.

The analysis involved preparing the data for binary classification, transforming textual input into numerical representations, training classification models, and evaluating predictive performance using appropriate metrics.

3.2.1 Feature Engineering: Text Vectorization

Textual data were processed using a Spark ML pipeline consisting of tokenization, stopword removal, TF–IDF feature extraction, and label encoding. Additionally, Word2Vec embeddings were explored to capture semantic relationships within the text.

The dataset was partitioned into training and testing subsets using an 80/20 split with a fixed random seed to ensure reproducibility and consistent evaluation.

3.2.2 Baseline Model: Logistic Regression with TF–IDF

A Logistic Regression classifier was trained on the training subset to predict the binary sentiment label using TF–IDF feature representations. Model performance was evaluated on the held-out test set. The resulting performance metrics were:

Metric	Value
Accuracy	0.8371
F1 Score	0.8566
Weighted Precision	0.9003
Weighted Recall	0.8371

Table 3.3: Model Performance Metrics

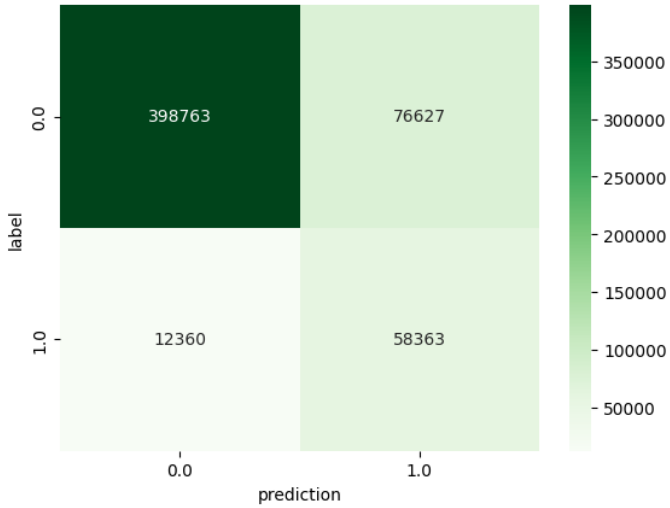

Figure 3.4: Confusion Matrix

These values confirm that TF–IDF remains a highly effective representation for large-scale sentiment classification. Logistic Regression also remains an appropriate baseline due to its computational efficiency, interpretability, and strong performance on sparse textual vectors.

However, the training data remained highly imbalanced, the model benefited from the dominance of the positive class. As a consequence, its metrics likely overstate its ability to distinguish negative reviews fairly. The baseline therefore remains useful as a reference model, but not as the most reliable indicator of balanced performance. To reduce the distortion introduced by imbalance, the notebook evaluated models trained on balanced datasets using Word2Vec embeddings. Two balancing approaches were considered: undersampling and controlled oversampling.

Undersampling. Logistic Regression with Word2Vec under undersampling achieved an accuracy and F1-score of **0.7667**, while Gradient-Boosted Trees under the same condition performed worse, with both accuracy and F1-score equal to **0.6919**. These results indicate that although class balance was achieved, too much information was discarded in the process. This is particularly problematic for models relying on semantic embeddings, since reducing the dataset also weakens the quality of the learned representation.

Oversampling. In this case, the minority class was replicated while the majority class was sampled down to a manageable size, creating a balanced dataset without sacrificing as much information as naive undersampling. Under these conditions, Logistic Regression with tuned Word2Vec features achieved an accuracy and F1-score of **0.7608**, while the tuned Gradient-Boosted Trees model achieved **0.7751** for both accuracy and F1-score. Among the balanced Word2Vec-based models, this was the best result.

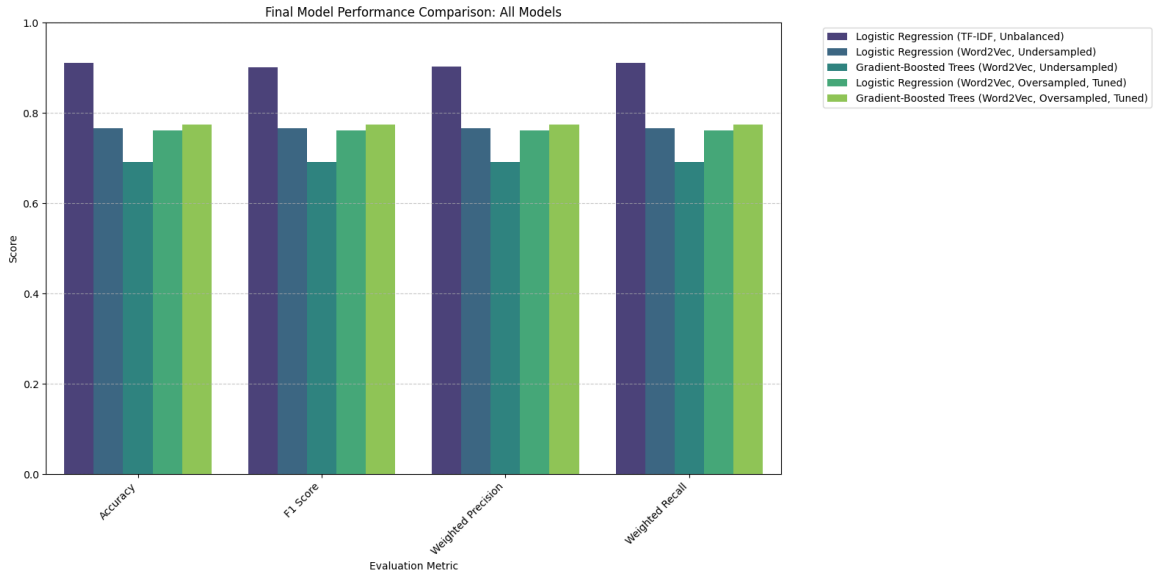

Figure 3.5: Performance comparison of sentiment classification models under different feature representations and balancing strategies.

3.2.3 Conclusion

The baseline TF–IDF + Logistic Regression model appears to outperform all alternative configurations, achieving the highest overall Accuracy and F1-score. However, this result must be interpreted with caution as heavily influenced by majority-class performance.

The undersampled experiments resulted in substantial loss of training data. The reduction in available examples negatively impacted model performance, as reflected in the lower evaluation metrics for Logistic Regression (Word2Vec, undersampled) and Gradient-Boosted Trees (Word2Vec, undersampled) compared to the baseline TF–IDF model.

The oversampled setting produced more reliable results. Both Logistic Regression and Gradient-Boosted Trees exhibited improved performance compared to their undersampled counterparts.

These findings suggest that Gradient-Boosted Trees can exploit the semantic structure of Word2Vec more effectively than Logistic Regression, but only when the training set remains sufficiently large. The apparent superiority of the tuned GBT model is therefore not simply due to the classifier itself, but to the combination of non-linear modeling, semantic embeddings, and a more appropriate balancing strategy.

3.3 Logistic Regression with N-Grams and Class Weights

N-grams are contiguous sequences of N words extracted from text. For example, unigrams correspond to single words, bigrams to pairs of consecutive words, and trigrams to sequences of three consecutive words. In sentiment classification, N-gram features are useful because they capture short contextual patterns such as *very good* or *not recommended*, which may convey sentiment more effectively than individual words alone.

The text processing pipeline included tokenization, stopword removal, bigram generation through the NGram transformer, HashingTF, IDF scaling, and label indexing. The class weights were computed as **0.5739** for the positive class and **3.8823** for the negative class.

The final model achieved an accuracy of **0.8735**, an F1-score of **0.8837**, a weighted precision of **0.9030**, a weighted recall of **0.8735**, and a mean absolute error of **0.1265**.

This shows that the classifier is considerably better at avoiding incorrect positive predictions, which strengthens its reliability in practical sentiment classification.

Although the comparison model achieves slightly fewer false negatives and therefore somewhat higher

recall for the positive class, this improvement comes at the cost of a much larger number of false positives. By contrast, the N-gram model achieves a more favorable trade-off between precision and recall, maintaining strong detection of positive reviews while significantly improving discrimination of the negative class. This suggests that the richer contextual information captured by N-gram features allows the model to better distinguish sentiment patterns than simpler text representations.

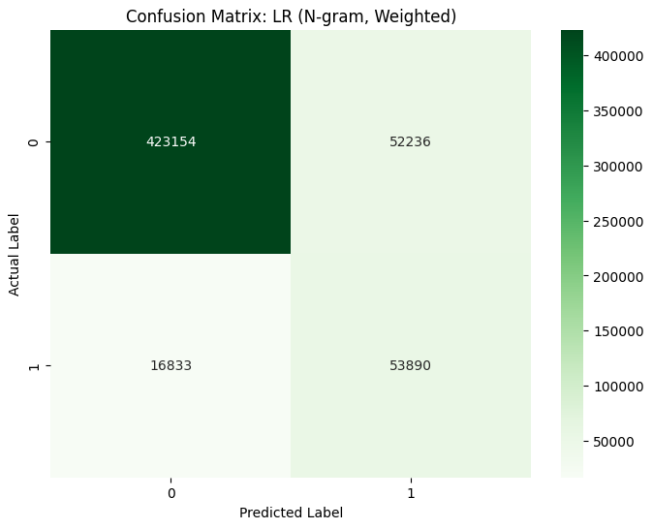

Figure 3.6: Confusion matrix of class-weighted Logistic Regression with N-gram features.

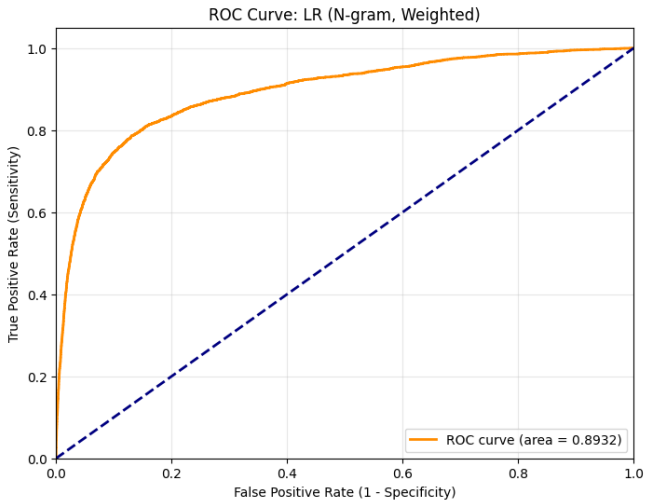

Figure 3.7: ROC curve of class-weighted Logistic Regression with N-gram features.

Conclusion. Overall, the N-gram Logistic Regression model should be considered the preferable configuration. Its higher accuracy, lower false positive rate, and more balanced confusion matrix indicate that it offers a more robust and realistic sentiment classification performance. These results confirm that incorporating contextual bigram features, together with class weighting, meaningfully improves the model's ability to classify imbalanced review data. However, the remaining errors on the minority class indicate that further gains may require richer semantic representations or more expressive nonlinear models.

3.4 Limitations

Despite the robustness of the distributed pipeline and modeling framework, several limitations must be acknowledged.

First, the dataset exhibits strong positivity bias. The majority of reviews are rated 4 or 5 stars, reflecting a well-documented phenomenon in online review ecosystems where satisfied users are more likely to leave feedback. This imbalance influences both exploratory statistics and classification outcomes, potentially inflating evaluation metrics such as accuracy.

Second, the binary sentiment formulation simplifies the rating spectrum by excluding neutral reviews (rating = 3). While this design choice improves class separability, it reduces the granularity of sentiment representation and may overlook nuanced evaluative patterns.

Third, metadata retrieved via the Google Books API contains substantial missingness in variables such as ratingsCount and publisher. Feature removal was necessary to preserve data integrity, but this may have reduced the richness of predictive information.

Fourth, textual models based on TF–IDF discard word order and contextual relationships. Although Word2Vec addresses semantic similarity, it still relies on locally trained embeddings rather than large-scale pretrained language models. Consequently, subtle contextual or sarcastic expressions may not be fully captured.

3.5 Conclusion

This project demonstrated the application of distributed data mining techniques to a large-scale review dataset combining structured metadata and unstructured textual content. By leveraging Apache Spark, the analysis was conducted efficiently across approximately three million reviews of the Amazon Books Reviews dataset, highlighting the practical importance of scalable computation in real-world data science workflows.

Across the exploratory and predictive stages of the analysis, the results consistently showed that structural variables such as price, genre, and review volume explain only a limited share of rating behavior. By contrast, textual features provide substantially richer predictive information, particularly when sentiment is represented through context-sensitive lexical patterns.

The modeling experiments further demonstrated that performance depends not only on classifier choice, but also on the interaction between feature representation and imbalance handling. The strong raw performance of the unbalanced TF–IDF baseline must be interpreted cautiously, since it partly reflects majority-class dominance. In contrast, the class-weighted N-gram Logistic Regression model and the oversampled, tuned Gradient-Boosted Trees model provided more balanced and methodologically reliable results. These findings confirm that the treatment of class imbalance is not a secondary preprocessing detail, but a main determinant of model validity.

The project illustrates an important principle in large-scale text mining: robust predictive performance emerges not simply from increasing model complexity, but from aligning representation, data preparation, and evaluation strategy with the structure of the problem. The strongest models were those that preserved as much information as possible while explicitly addressing imbalance and contextual variation in language. Overall, the project demonstrates how distributed data mining methods can be used not only to scale analysis to millions of records, but also to uncover the interplay between user behavior, writing style, and perceived quality in digital review ecosystems.